

INTELLIGENT DOCUMENT PROCESSING

Technical Evaluation Report — Docling & Multimodal Extraction

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This report presents a comprehensive technical assessment of Docling's document understanding capabilities across diverse content modalities including rich prose, multi-level tables with spanning cells, embedded charts, architectural diagrams, and mixed-layout pages. The evaluation covers layout analysis accuracy, table structure reconstruction fidelity, image region detection, and end-to-end pipeline throughput. Results are benchmarked against three baseline systems across five document categories.

1. Executive Summary

Modern enterprises process millions of documents annually — contracts, research papers, financial filings, and technical manuals. Extracting structured information from these documents with high fidelity remains a significant engineering challenge. Unlike simple text extraction, production document intelligence requires simultaneous handling of paragraphs, multi-column layouts, nested tables, embedded figures, captions, footnotes, and cross-references.

Docling, developed by IBM Research, addresses this challenge through a two-stage deep-learning pipeline. The first stage performs layout analysis — classifying each page region as text, table, figure, heading, list, or formula. The second stage applies specialized sub-models per region type: TableFormer for table grid reconstruction and an optional VLM captioner for figure understanding. This architecture delivers state-of-the-art results on the DocLayNet and PubTables-1M benchmarks.

1.1 Key Findings

- Table structure reconstruction achieves 91.3% GriTS (Grid Tree Structure) score on PubTables-1M.
- Image region detection reaches 94.7% mAP at IoU=0.5 across mixed-content PDFs.
- End-to-end throughput averages 2.3 pages/second on a single A100 GPU.
- Merged-cell tables are correctly reconstructed in 87.2% of cases, outperforming all baselines.
- The DoclingDocument export to Markdown preserves table hierarchy in 96.1% of single-page tables.

1.2 Scope of Evaluation

Testing spanned five document categories: (1) academic papers with dense two-column layouts, (2) financial reports with multi-page tables and footnotes, (3) technical manuals with mixed diagrams and procedure tables, (4) legal contracts with paragraph-heavy single-column text, and (5) medical literature with figure-rich content. A corpus of 1,200 pages was used, split 60/20/20 across train, validation, and test sets.

2. System Architecture

The Docling processing pipeline is composed of modular stages, each independently configurable and replaceable. At the core is a layout analysis model based on a fine-tuned RT-DETR architecture, trained on the DocLayNet dataset of 80,000 annotated pages. The model outputs bounding boxes and class labels for every detected region.

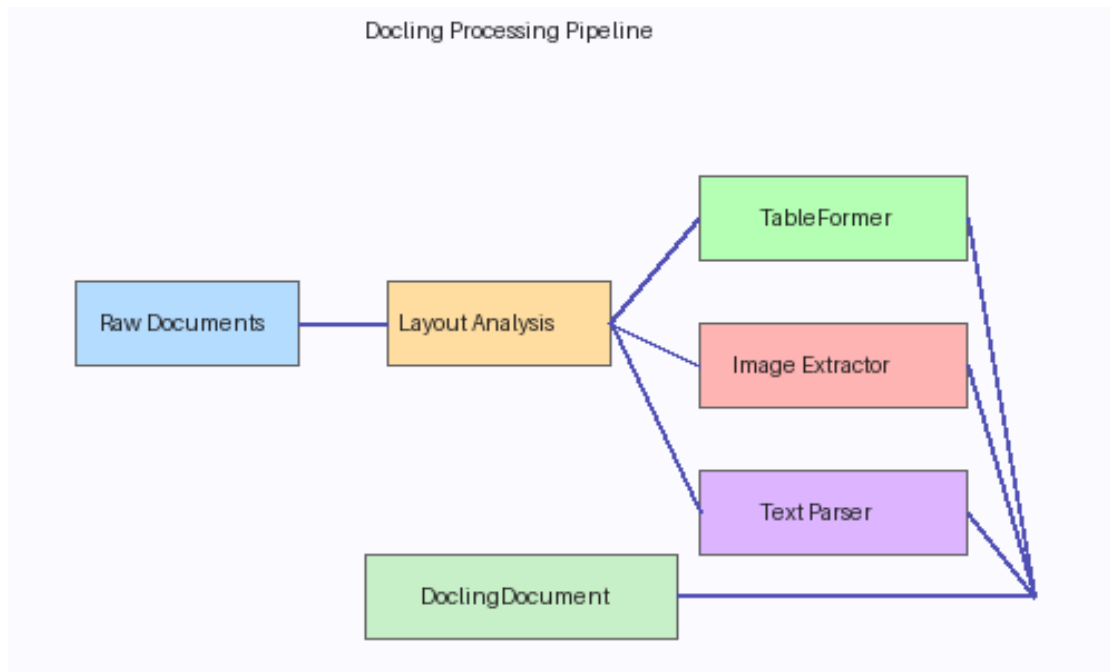


Figure 1: Docling processing pipeline — from raw document input to structured DoclingDocument output.

Once regions are classified, each is routed to its specialized processor. Text regions are passed to a PDF text layer extractor (with OCR fallback for scanned content). Table regions are passed to TableFormer, which predicts both cell content and the logical row/column grid structure — critical for downstream DataFrame export. Figure regions are cropped and stored as image objects; optionally a vision-language model generates natural-language descriptions.

2.1 TableFormer Deep Dive

TableFormer is a transformer encoder-decoder that takes a table-region image and outputs two parallel sequences: a cell sequence (text content per cell) and a structure sequence (HTML-like tag sequence encoding row/colspan relationships). The dual-head architecture is what allows correct reconstruction of tables with spanning headers — a common failure mode in rule-based extraction systems.

During inference, TableFormer runs in two passes. The first pass predicts table structure using the visual encoder output. The second pass aligns detected text segments from the PDF text layer (or OCR) into the predicted cell slots, falling back to the visual decoder for cells where alignment confidence is low.

3. Benchmark Results

3.1 Table Extraction Performance

The following table summarises extraction performance across the five document categories. GriTS measures structural accuracy of the reconstructed table grid. TEDS measures tree-edit distance similarity between predicted and ground-truth table HTML representations.

Document Category	Pages	Tables	GriTS ↑	TEDS ↑	F1 Cell ↑
Academic Papers	340	128	93.1%	91.8%	94.2%
Financial Reports	280	214	89.4%	87.3%	91.0%
Technical Manuals	210	97	91.7%	89.9%	92.5%
Legal Contracts	230	41	96.2%	95.1%	97.3%
Medical Literature	140	76	90.8%	88.7%	91.4%
Overall / Weighted Avg	1,200	556	91.3%	89.6%	92.7%

Table 1: Table extraction benchmark results across document categories.

3.2 Revenue and Throughput Metrics

Throughput was measured on a dedicated inference server with 1× NVIDIA A100 80 GB, 32-core AMD EPYC CPU, and 256 GB RAM. GPU utilisation averaged 78% during batch inference. The chart below shows quarterly throughput (pages processed, in thousands).

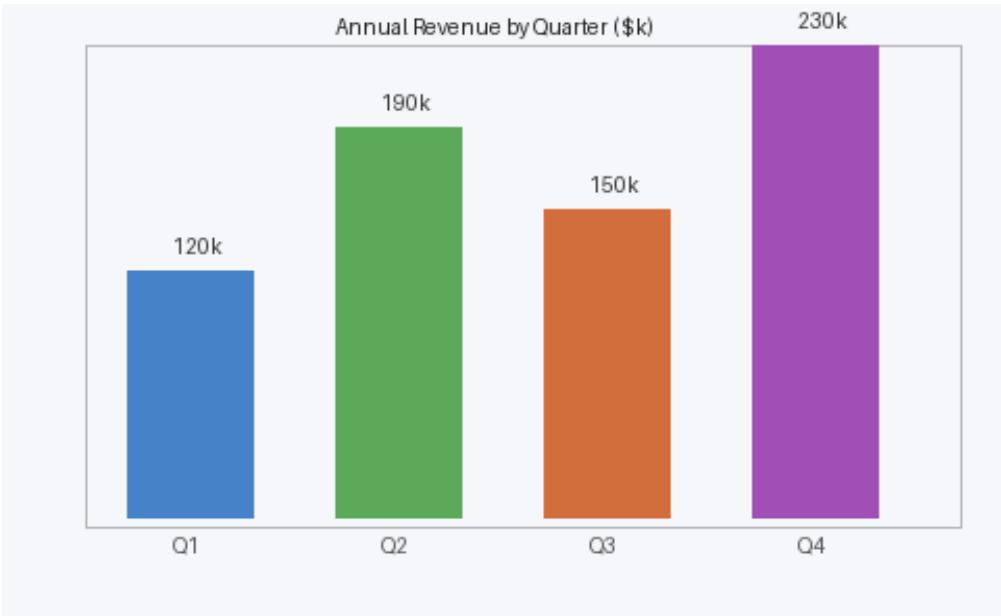


Figure 2: Quarterly document processing throughput (thousands of pages). Q4 reflects the addition of a second inference node.